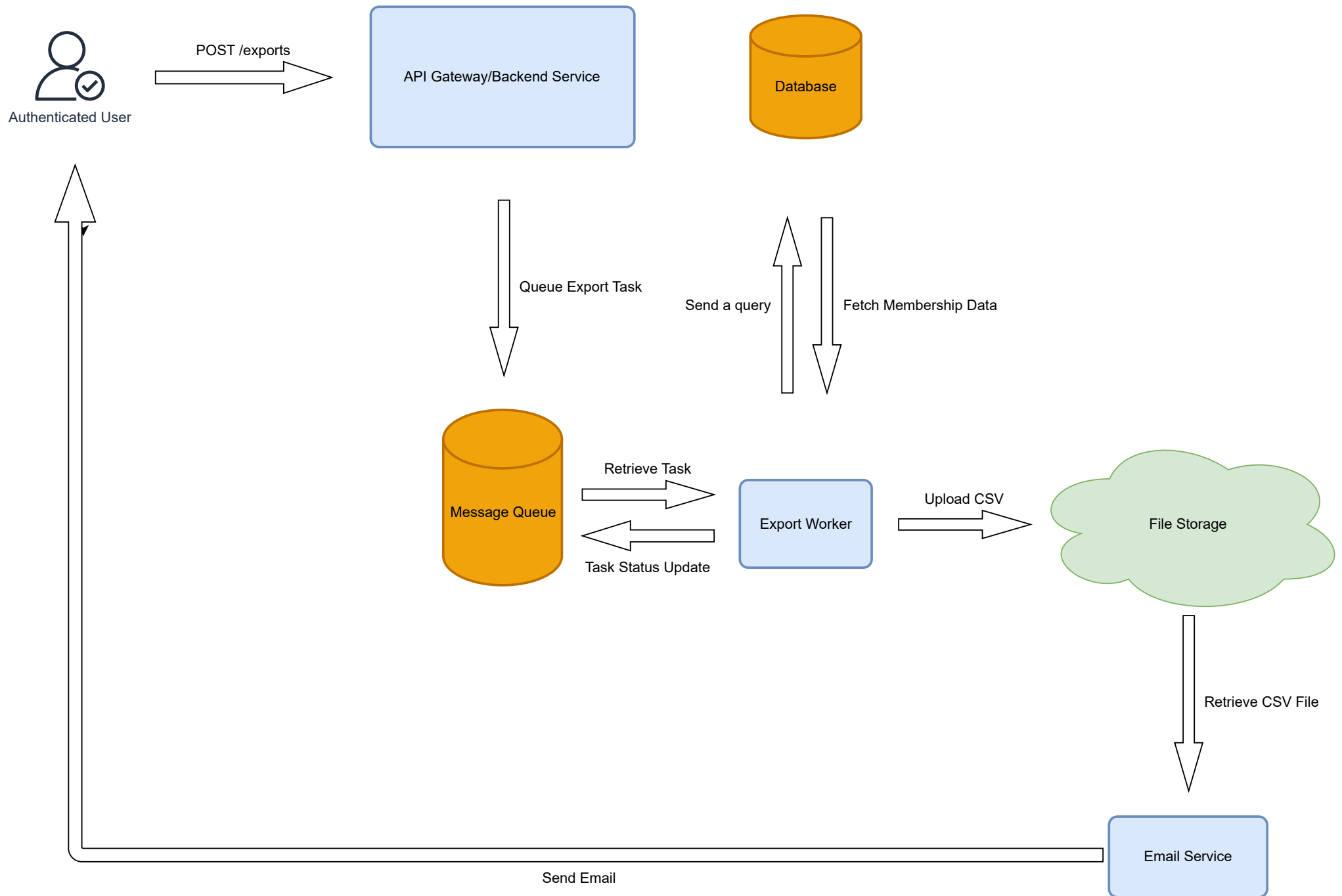




Assumptions and Design Decisions

1. **Asynchronous Process:**

Export tasks are handled asynchronously using a message queue (RabbitMQ) to avoid blocking the API during CSV generation

2. **Scalability:**

RabbitMQ is used to decouple the API from the Export Worker, allowing for horizontal scaling of workers if demand increases

3. **Fault Tolerance:**

The Message Queue ensures task reliability, with retry mechanisms for failed exports and dead-letter queues for error handling

4. **Security:**

Only authenticated users can initiate export tasks. The generated CSV files are stored securely in AWS S3 with time-limited access URLs

5. **File Storage:**

AWS S3 is used for file storage to ensure durability, availability, and cost-effectiveness

6. **Extensibility:**

The architecture allows for future integration with additional file formats (e.g., PDF) or export triggers (e.g., scheduled exports)

7. **Database Queries:**

Membership data is fetched from already existing database using optimized queries to ensure minimal latency

8. **Email Delivery:**

SendGrid is chosen as the email service provider for reliable and scalable delivery of export notifications

9. **Monitoring and Logging:**

Logs are recorded at each step of the process for debugging and performance monitoring

This design ensures a stable, scalable, and maintainable solution for exporting membership data.